Chapter:4
Three Phase Induction motor

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Introduction

- The **three phase induction motors** are the most widely used electric motors in the industry.
- They run at **essentially constant speed** from no-load to full-load.
- Induction motor runs at a speed other than the synchronous speed, so it is called asynchronous motor.
- Nevertheless, these, **motors** are simple, rugged, low-priced, easy to maintain and can be manufactured with characteristics to suit most industrial requirements
- It works on the principle of electromagnetic induction.
- It has a **stator and a rotor**.
- The ***induction motor* may be considered as a transformer** with a rotating secondary and it can, therefore, be described as a "transformer type" ac machine in which electrical energy is converted into mechanical energy.

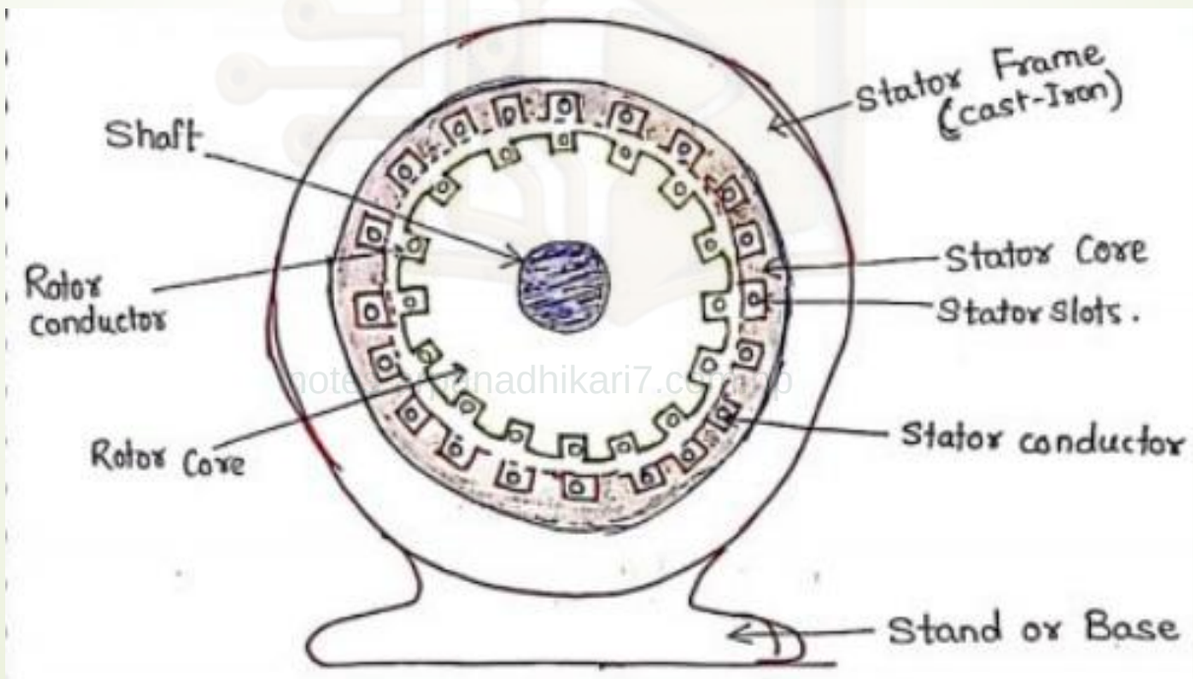
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Construction of three phase induction motor

A three-phase induction motor consists of two main parts namely,

- Stator
- Rotor

There is **a small air gap** between the stator and rotor which ranges **from 0.4 mm to 4** mm depending on the power rating of the motor.



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Stator

The stator of a three-phase induction motor is a stationary part.

It consists of following parts,

1. Stator frame: It is made up of **cast-iron and holds the stator core**.
2. Stator core: It is made up of **thin sheets of steel lamination** stacked together. Stator core has **slots** for providing three-phase ac windings in it.
3. Stator winding(three phase winding): These windings consists of **insulated copper wire**. The **three phase** are connected in either star or delta fashion. Stator winding are done for specific **number of poles, as per requirement.**

$$Ns = 120f/p$$

- When a 3 phase ac supply is given to the stator winding a r**otating magnetic field** is set up in space
- Thus the rotating magnetic field **induces emf** in the rotor by electromagnetic induction, that **produces torque** , resulting in rotation of motor.

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- The air gap between the stator and rotor needs to be small as practically possible.

1. **It reduces the leakage flux between stator and rotor**
2. **It improves the operating power factor of induction motor.**

Rotor

- The rotor is a **rotating or moving part** of the three-phase induction motor.
- It consists of a rotor core made up of thin **laminations of high grade silicon steel** to reduce the hysteresis and eddy-current losses. The rotor core is a hollow cylinder, mounted on a shaft. On outer periphery of the rotor core, **slots** are provided to place the rotor winding.

Based on the construction, the rotor of a three-phase induction motor can be of the following two types −

1. Squirrel-cage rotor
2. Wound rotor

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Difference between Squirrel cage and Wound rotor

Basis For Comparison	Slip Ring Motor	Squirrel Cage motor
Defintion	The rotor of the motor is constructed as a slip ring type.	The rotor of the motor is a squirrel cage type.
Rotor	Cylindrical laminated core with parallel slots and each slot consist one bar.	The slots of the rotor are not parallel, but are skewed.
Other name	Phase wound rotor	Cage motor
Construction	Complicated	Simple
Resistance	Added external to the rotor	The rotor bar is permanently shorted at the end of the ring, thus it is not possible to add any external resistance.
Starter	The rotor resistance starter can be used.	Rotor resistance starter can not be used.

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Starting Torque	High	Low
Brushes	Present	Absent
Maintenance	Frequent maintenance required	Less maintenance required
Copper Loss	High	Low
Efficieny	Low	High
Speed Control	Possible	Not Possible
Power Factor	Low	HIgh
Cost	Costly	Cheap
Starting Current	Low	High
Uses	Use in hoist, cranes, elevator where high torque is required.	Use in lathe machines, fan, blower, profiting machines, etc.

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Principle of operation of induction motor

- Induction motor works on principle of **electromagnetic induction.**
- When 3 phase supply is given to 3 phase stator winding, a **rotating magnetic field is created, along with stator flux**
- The rotating magnetic field cuts the **rotor** conductor, **inducing an EMF** in it as per **Faradys law of electromagnetic induction.**
- The induced EMF causes a **current to flow in the rotor** as rotor circuit is short circuited.
- The **induced current in rotor** will also produce **alternating flux** around it. Besides, **rotor flux lags behind the stator flux as dictated by Lenz law**(rotor flux will tend to oppose the cause of producing it i.e. stator flux.)
- Here, the cause of **production** of **rotor current** is the **relative velocity between rotating stator flux and the rotor,**
- The rotor will try to **catch up** with the **stator rotating magnetic field.** Thus the **rotor rotates in the same direction** as that of stator flux to **minimize the relative velocity**.
- However, the rotor **never succeeds** in catching up the **synchronous speed**. This is the **basic working principle of induction motor** of either type, single phase of 3 phase.

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- The rotational speed of the **rotating magnetic field** is called as **synchronous speed.**

$$Ns = \frac{120 \times f}{P} \quad (RPM)$$

where, f = frequency of the supply

P = number of poles

Slip:

- **Roto**r tries to catch up the **synchronous speed** of the stator field, and hence it **rotates**.
- But in practice, **rotor never succeeds** in catching up.
- If **rotor catches up the stator speed**, there **wont be any relative speed** between the stator flux and the rotor, hence **no induced rotor current** and **no torque** production to maintain the rotation.
- That is why the **rotor rotates** at speed which is always **less the synchronous speed.**
- The difference between the synchronous speed (N_s) and actual speed (N) of the rotor is called as slip.

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$$\% \text{ slip } \quad s = \frac{Ns - N}{Ns} \times 100$$

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Applications of 3-Phase Induction Motor

- A three-phase induction motor might be used to power a centrifugal pump that moves crude oil from wells to pipelines or storage tanks.
- It is used to drive a compressor that pressurizes natural gas for transportation or storage.
- It is used to run a turbine that generates electricity for offshore platforms or remote locations.
- It is used to power a pump that moves liquids through distillation columns or catalytic crackers.
- It is used to drive a compressor that increases the pressure of gases for further processing or separation.
- It is used to drive a transformer that converts high voltage from power plants to lower voltage for transmission lines or vice versa.
- It is also used to operate a switchgear that connects or disconnects circuits for protection or control purposes.
- Induction motors are versatile and widely used AC electric motors that operate on the principle of electromagnetic induction.
- They have different types, such as squirrel cages or slip rings, and different applications, such as pumps, fans, compressors, conveyors, cranes, elevators, and more.

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